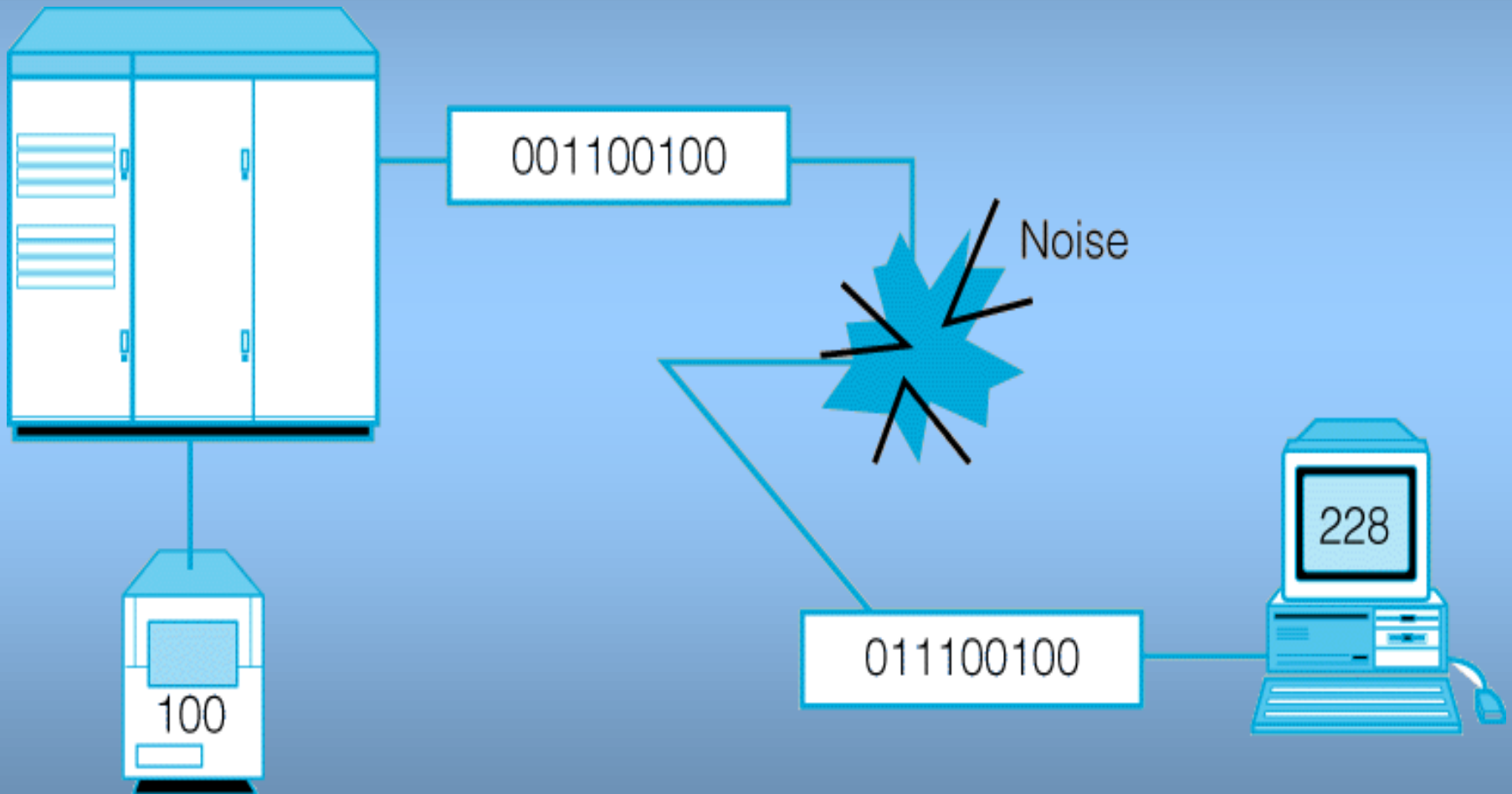


Transmission Errors 1

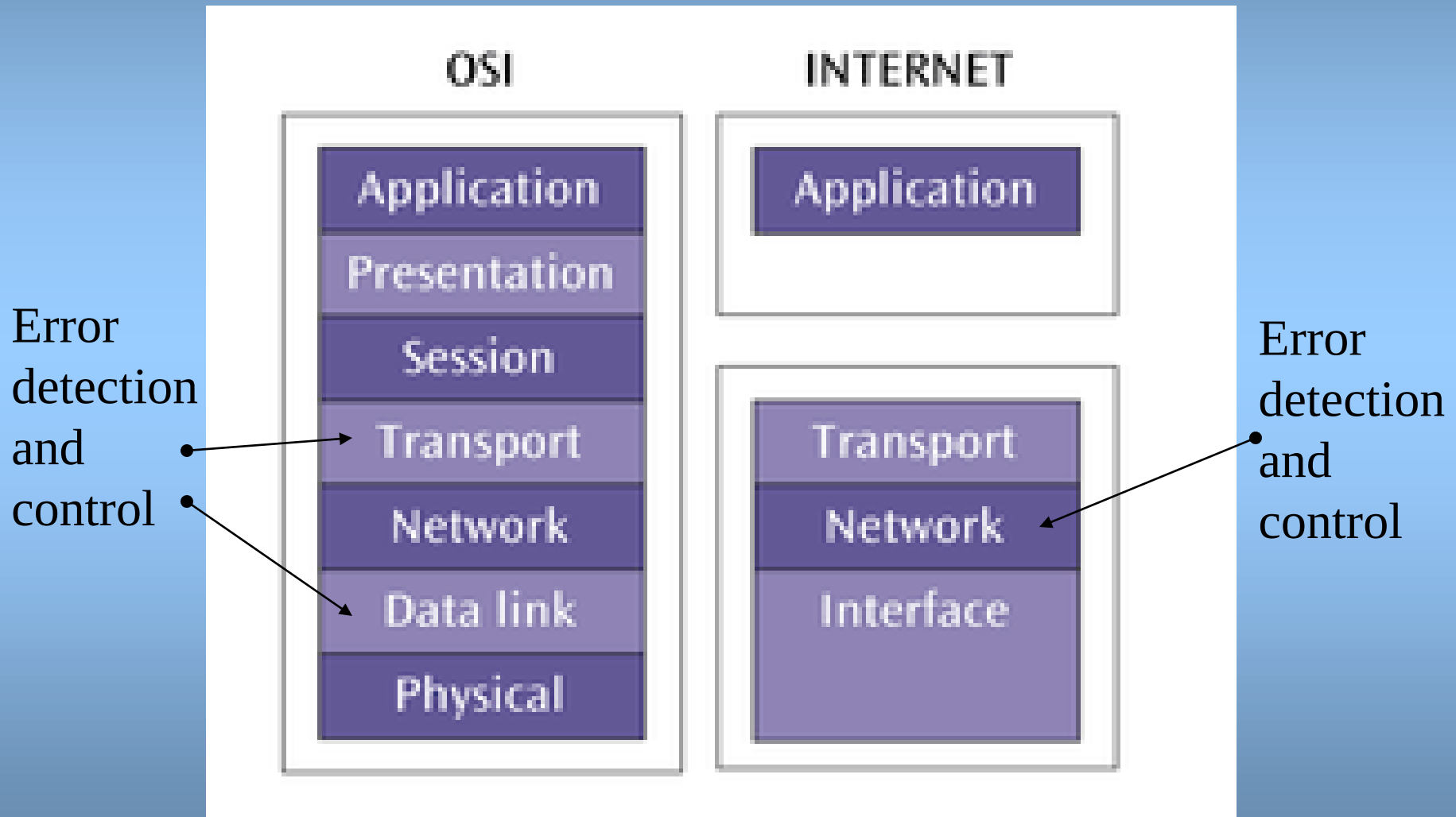
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University of Moratuwa

**Reference: Data Communications and Computer Networks: A Business User's
Approach by Curt M. White**

Transmission Error



OSI vs Internet Model



Error detection in different layers

- Some error detection in other layers
 - Transport layer in OSI
 - Network layer in Internet model
- Focus on the data link layer
 - Independent of the medium or physical layer

Data Link Layer

- Responsible for taking the data and transforming it into a *frame* with header, control and address information.
- Physical path communication
- Error detection
- Error correction
- Resolve competing requests

Overview

- Errors Happen
- Noise and Errors
- Error Prevention
- Error Detection Techniques
- Error Control- Next week

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Chapter 6

Introduction

Noise is always present.

If a communications line experiences too much noise, the signal will be lost or corrupted.

Communication systems should check for transmission errors.

Once an error is detected, a system may perform some action.

Some systems perform no error control, but simply let the data in error be discarded.

Noise and Errors – White Noise

Also known as thermal or Gaussian noise

Relatively constant and can be reduced.

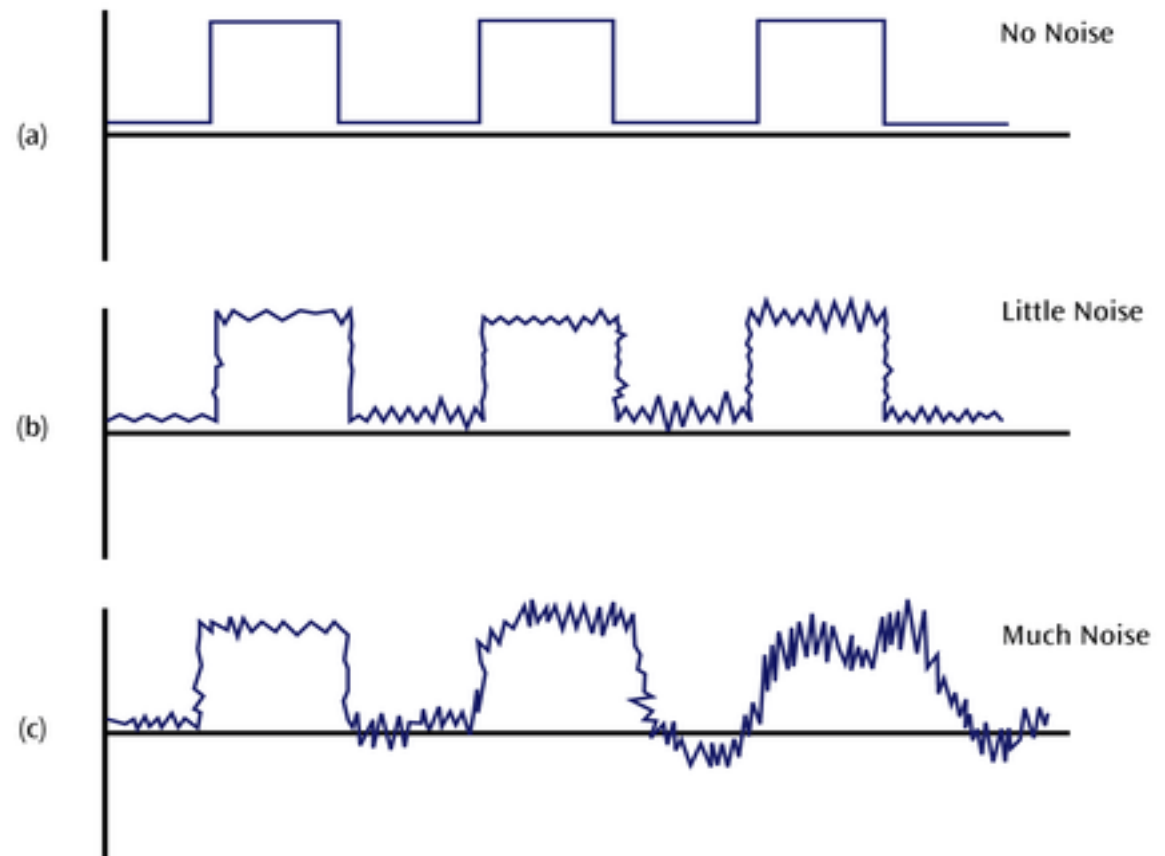
If white noise gets too strong, it can completely disrupt the signal.

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Figure 6-1

White noise as it interferes with a digital signal



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Noise and Errors – Impulse Noise

One of the most disruptive forms of noise.

Random spikes of power that can destroy one or more bits of information.

Difficult to remove from an analog signal because it may be hard to distinguish from the original signal.

Impulse noise can damage more bits if the bits are closer together (transmitted at a faster rate).

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Figure 6-2
*The affect of impulse
noise on a digital signal*

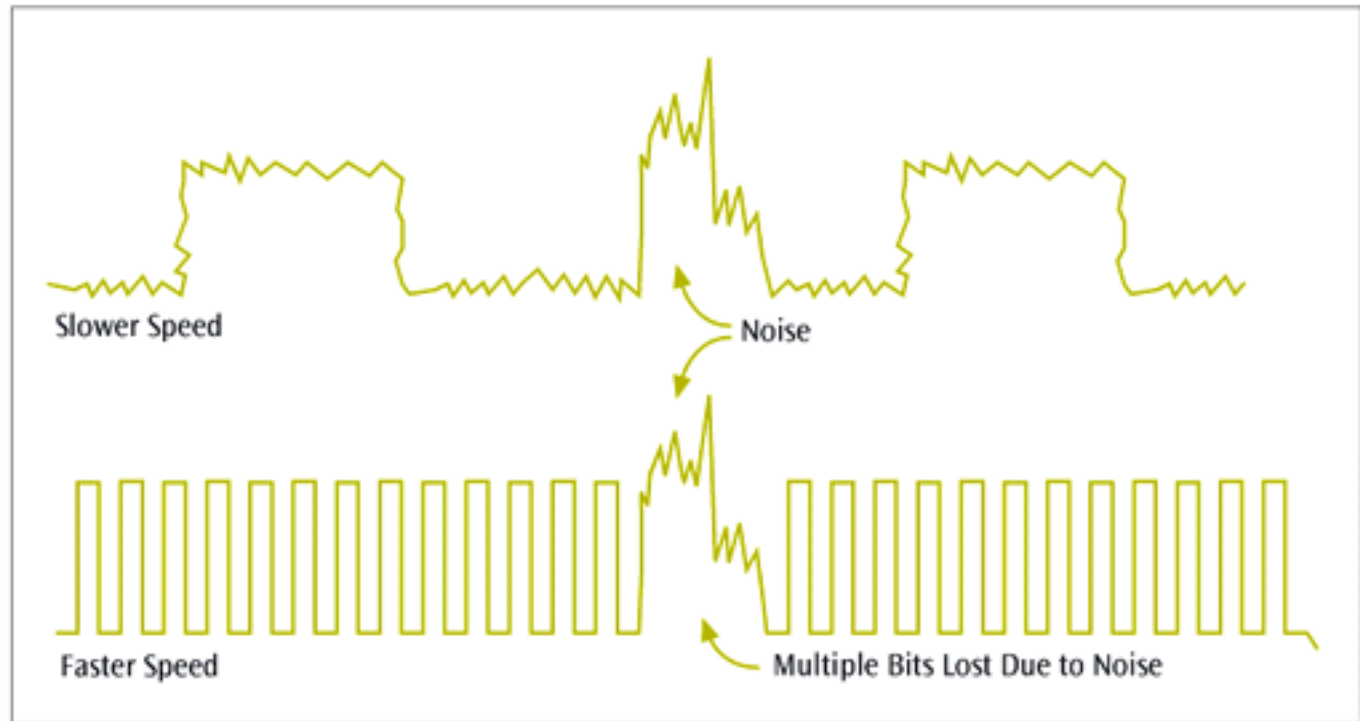


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Figure 6-3

*Transmission speed
and its relationship
to noise*



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Noise and Errors - Crosstalk

Unwanted coupling between two different signal paths.

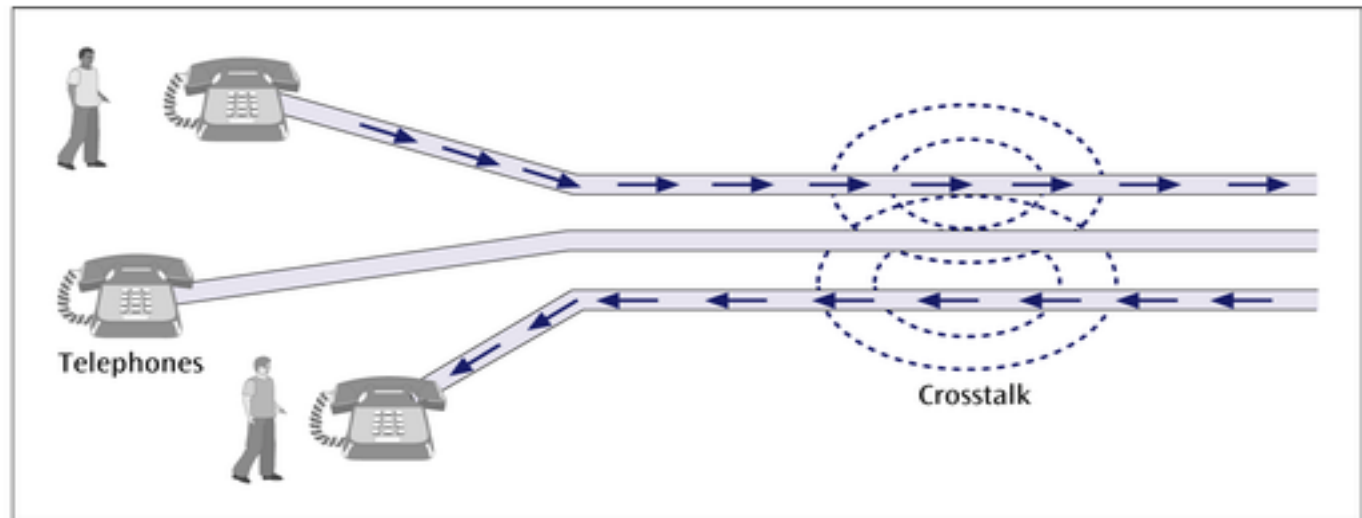
For example, hearing another conversation while talking on the telephone.

Relatively constant and can be reduced with proper measures.

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Figure 6-4
*Two telephone circuits
experiencing crosstalk*



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Noise and Errors - Echo

The reflective feedback of a transmitted signal as the signal moves through a medium.

Most often occurs on coaxial cable.

If echo bad enough, it could interfere with original signal.

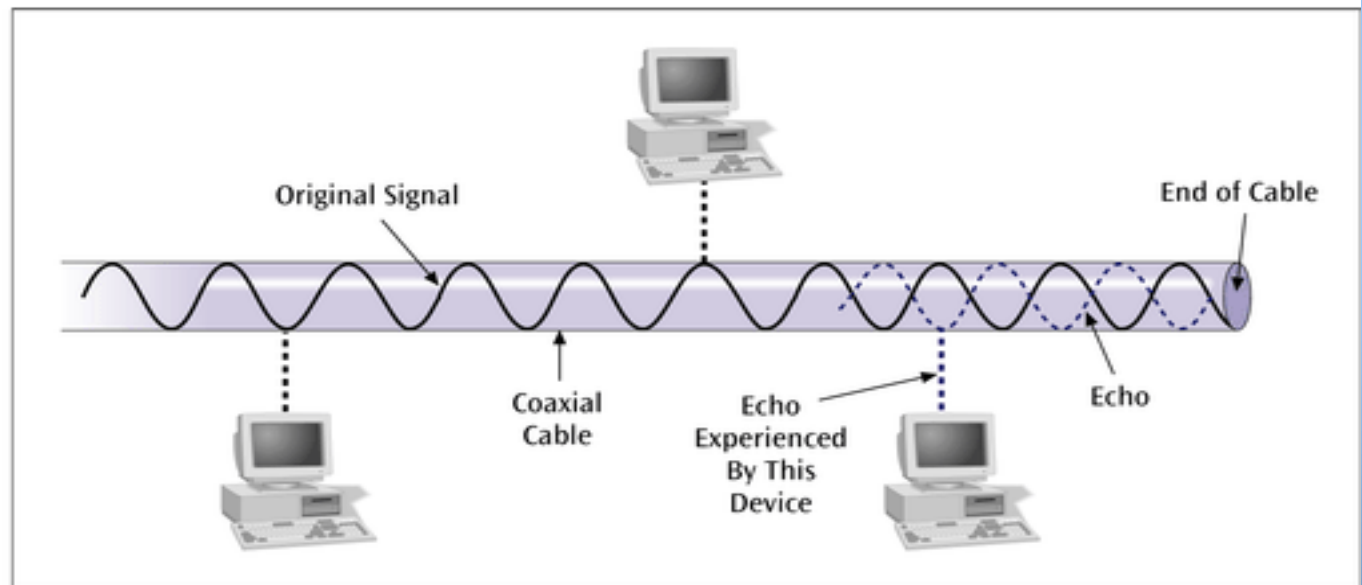
Relatively constant, and can be significantly reduced.

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Figure 6-5

A signal bouncing back at the end of a cable and causing echo



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Noise and Errors - Jitter

The result of small timing irregularities during the transmission of digital signals.

Occurs when a digital signal is repeated over and over.

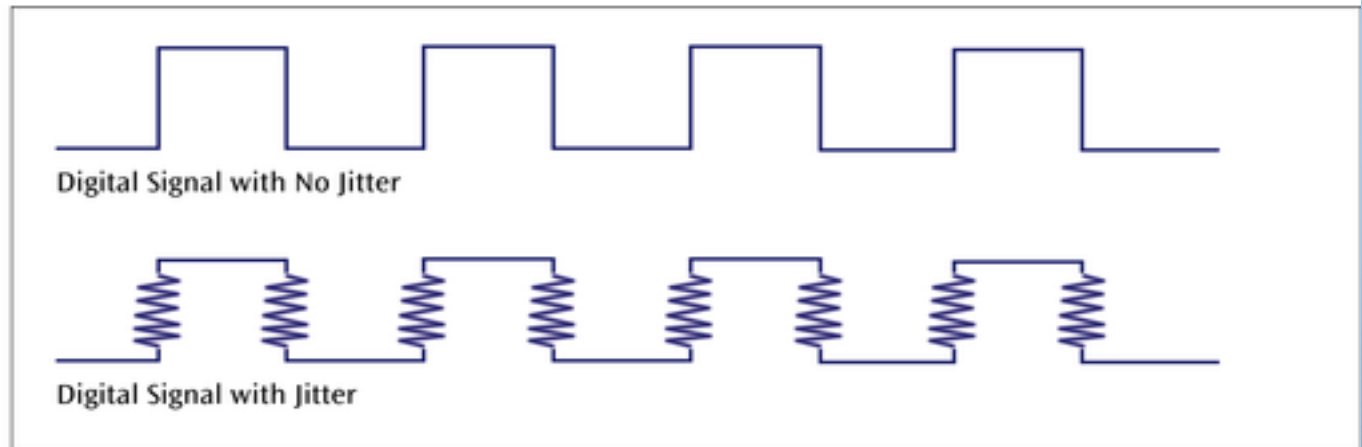
If serious enough, jitter forces systems to slow down their transmission.

Reduce jitter - shielding

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Figure 6-6
*Original digital signal
and digital signal
with jitter*



Noise and Errors – Delay Distortion

Occurs because the velocity of propagation of a signal through a medium varies with the frequency of the signal.
Can be reduced using equalizers

Attenuation

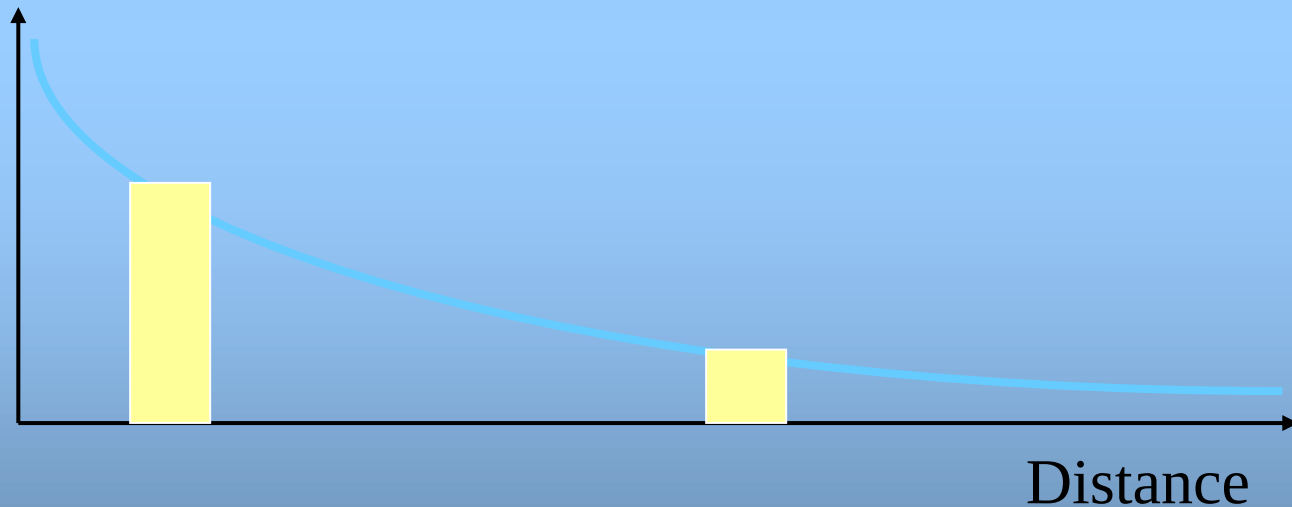
The continuous loss of a signal's strength as it travels through a medium.

Use less lossy medium

Use amplifiers

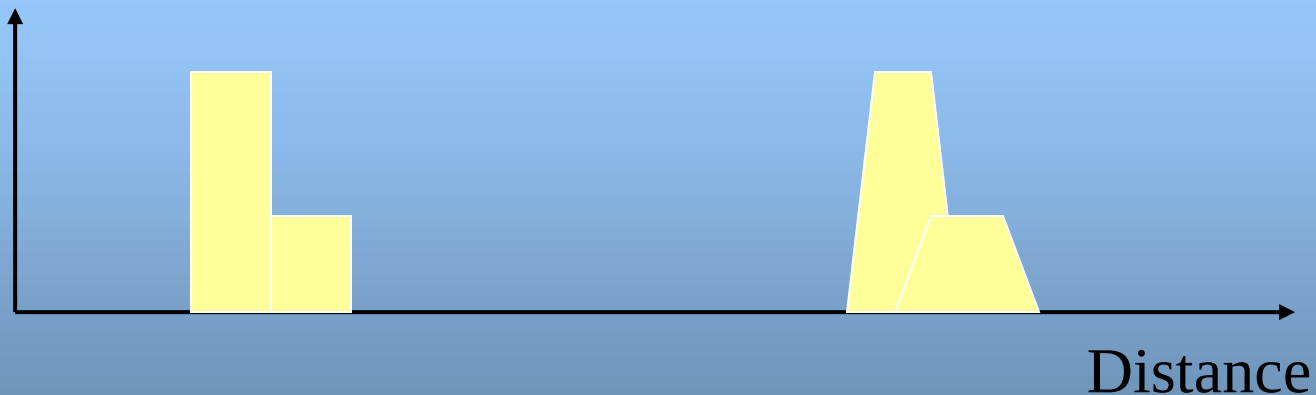
Attenuation

- When signal travels, it gets weaker.
 - If too weak, cannot tell 1s and 0s.



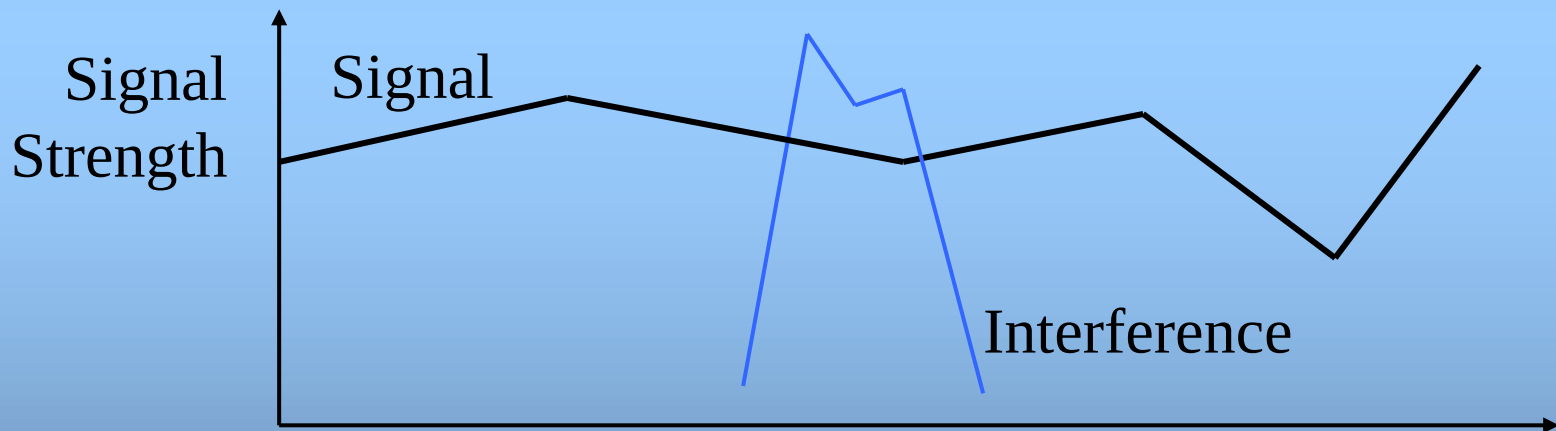
Distortion

- As signal travels, it become distorted.
 - Changes shape
 - Successive bits may merge, making reception difficult



Interference

- Unwanted signal from outside sources
 - Often intermittent, difficult to diagnose



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Error Prevention

To prevent errors from happening, several techniques may be applied:

- Proper shielding of cables to reduce interference
- Telephone line conditioning or equalization
- Replacing older media and equipment with new, possibly digital components
- Proper use of digital repeaters and analog amplifiers
- Observe the stated capacities of the media

CS2033: Data Communication and Networking

Transmission Errors 2

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Information Redundancy - Coding

- A data word with d bits is encoded into a codeword with c bits - $c > d$
- To extract original data - c bits must be decoded
- If the c bits do not constitute a valid codeword an error is detected
- For certain encoding schemes - some types of errors can also be corrected

Separability of a Code

- A code is separable if it has separate fields for the data and the code bits
- Decoding consists of disregarding the code bits
- The code bits can be processed separately to verify the correctness of the data
- A non-separable code has the data and code bits integrated together - extracting the data from the encoded word requires some processing

Parity Codes

- The simplest separable codes are the parity codes
- A parity-coded word includes d data bits and an extra bit which holds the parity
- In even parity code - the extra bit is set so that the total number of 1's in the $(d+1)$ -bit word (including the parity bit) is even
- In even odd parity code - the extra bit is set so that the total number of 1's in the $(d+1)$ -bit word (including the parity bit) is odd

Parity Code Example

- Even parity of 1s

Data							Parity
1	0	0	1	1	1	0	0
0	0	1	1	1	0	0	1
1	1	1	0	1	1	1	0

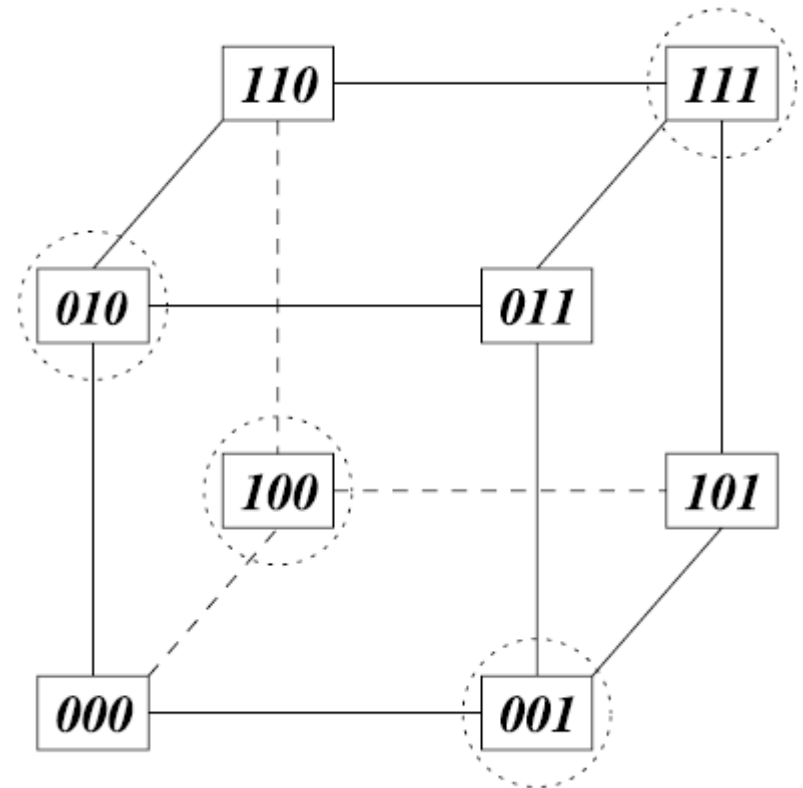
2-Dimensional Parity Example

- Even parity of 1s
- Longitudinal Redundancy Check(LRC)/Vertical Redundancy Check(VRC)
- Identifies a unique erroneous bit

Data							VRC
1	0	0	1	1	1	0	0
0	0	1	1	1	0	0	1
1	1	1	0	1	1	1	0
1	1	1	0	1	1	1	0
0	0	1	1	1	0	0	1
0	0	1	0	1	0	1	1
0	0	1	1	1	0	0	1
1	0	0	0	1	1	1	0
LRC							

Hamming Distance

- The Hamming distance between two codewords - the number of bit positions in which the two words differ
- Two words in this figure are connected by an edge if their Hamming distance is 1



Hamming Code

- Developed based on hamming distance concept
- Assigns multiple parity bits to cover each bit of data
- Many different hamming code schemes exist
- For simplicity we will only look at (7,4) Single Error Correcting (SEC) Hamming Code
 - 7-bit code (4 data bits + 3 parity bits)

(7,4) SEC Hamming Code

- Can correct 1-bit errors
- How to calculate number of parity bits?
 - $2^r \geq m + r + 1$
where r = number of parity bits
 m = message bits (data)
- If $m = 4$; according to above inequality $r = 3$

Parity bits placement

- In the resulting codeword, parity bits can be placed at different places. This placement affects how easy/difficult to detect and correct errors
- One way is to place parity bits at places corresponding to powers of two

$m_7m_6m_5p_4m_3p_2p_1$ ← Bit position

- E.g. say $m = 1001$
 $100p_41p_2p_1$

Parity bit coverage

7 6 5 4 3 2 1
1 0 0 P_4 1 P_2 P_1

Sender

	P_4	P_2	P_1
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

$P_1 \rightarrow 1, 3, 5, 7 \rightarrow P_1 101$

$P_2 \rightarrow 2, 3, 6, 7 \rightarrow P_2 101$

$P_4 \rightarrow 4, 5, 6, 7 \rightarrow P_4 001$

Assuming Even Parity;

$P_1 = 0$ $P_2 = 0$ $P_4 = 1$

Detecting Error

Receiving

1 0 0 1 1 0 0 ← original

Suppose received. 1 1 0 1 1 0 0

Compute parity;

$$P_1 \rightarrow 1, 3, 5, 7 \rightarrow 0 \oplus 0 \oplus 1 = 1$$

$$P_2 \rightarrow 2, 3, 6, 7 \rightarrow 0 \oplus 1 \oplus 0 \oplus 0 = 1$$

$$P_4 \rightarrow 4, 5, 6, 7 \rightarrow 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

	P_4	P_2	P_1
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Correcting Error

Received

1 0 0 1 1 0 0 $\leftarrow$ original

Suppose
received.

1 1 0 1 1 0 0

Compute parity;

$$P_1 \rightarrow 1, 3, 5, 7 \rightarrow 0101 = 0$$

$$P_2 \rightarrow 2, 3, 6, 7 \rightarrow 0111 = 1$$

$$P_4 \rightarrow 4, 5, 6, 7 \rightarrow 1011 = 1$$

$$P_4 P_2 P_1 = 110$$

decimal

6

* 6th bit is

the wrong bit.

Cyclic Redundancy Checksum (CRC)

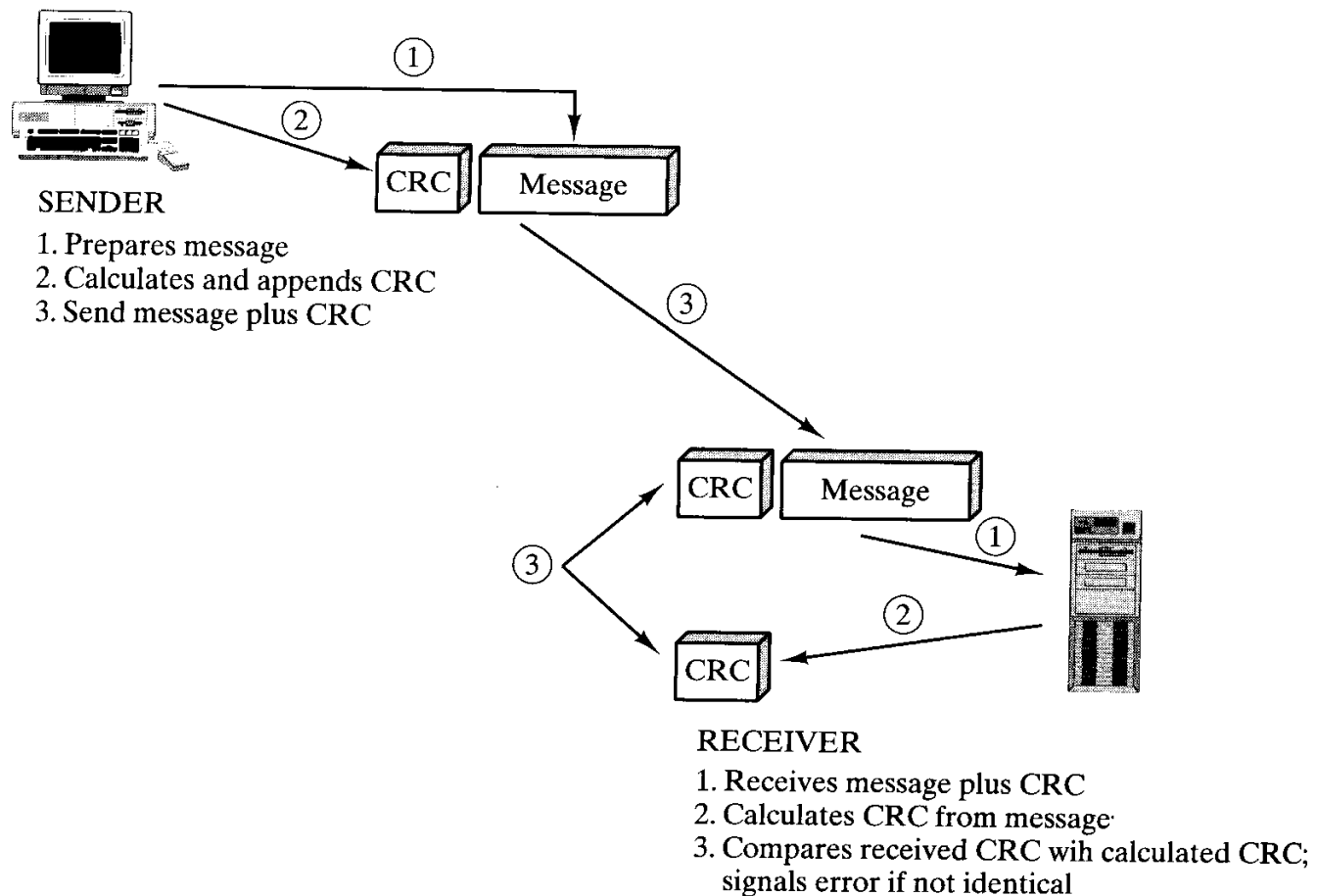
- A non-separable code
- The CRC error detection method treats the packet of data to be transmitted as a large polynomial.
- The transmitter takes the message polynomial and using polynomial arithmetic, divides it by a given generating polynomial.
- The quotient is discarded but the remainder is “attached” to the end of the message (remainder (mod) arithmetic)

Cyclic Redundancy Checksum (CRC)

- The message (with the remainder) is transmitted to the receiver.
- The receiver divides the message and remainder by the same generating polynomial.
- If a remainder not equal to zero results, there was an error during transmission.
- If a remainder of zero results, there was no error during transmission.

Cyclic Redundancy Checksum (CRC)

FIGURE 5.14 Error Detection Using CRC.



CRC Example

Message = 101101

CRC Polynomial = $x^3 + x^2 + 1$

CRC Polynomial bits = 1101

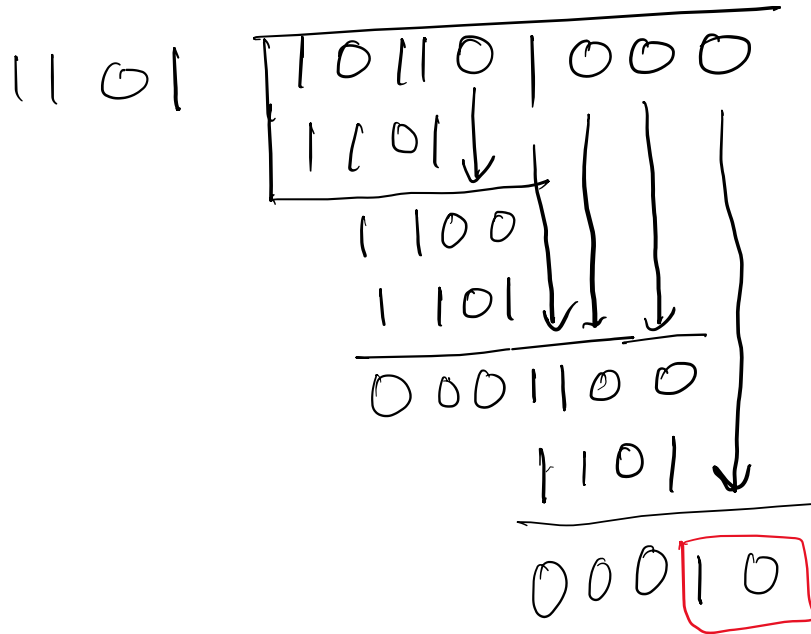
of CRC bits = 3 (1 bit less than polynomial bits)

How to calculate?

Modulo-2 long division (XOR)

Remainder is the CRC

Cyclic Redundancy Check



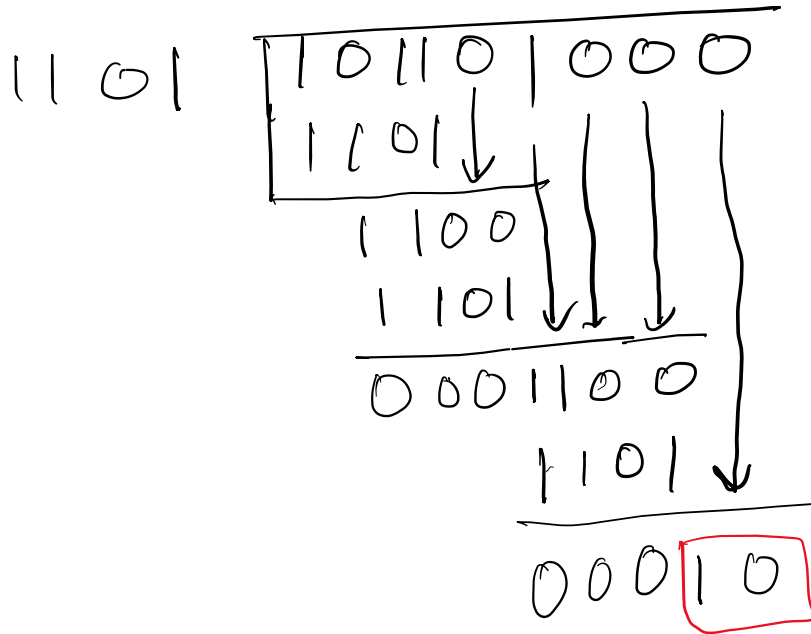
Remainder = 10

But we want 3 bits.

Thus $\Rightarrow$ 010 = CRC bits

Message + CRC is sent to other side.

Cyclic Redundancy Check



* At recipient,
divide Msg + CRC with
the same polynomial.
If remainder $\neq 0$,
ERROR

Remainder = 10

But we want 3 bits.

Thus $\Rightarrow$ 010 = CRC bits

Message + CRC is sent to other side.



Summary

- Information Coding
- Parity codes
- Hamming codes
- CRC